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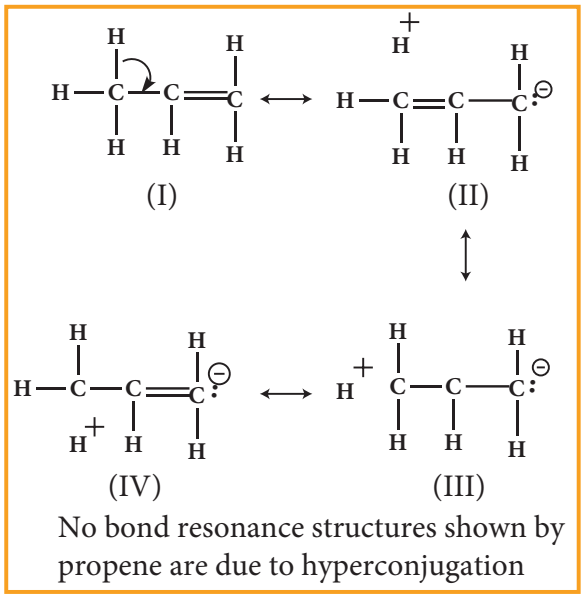
Hyper conjugation

The delocalisation of electrons of **σ** bond is called as hyper conjugation. It is a special stabilising effect that results due to the interaction of electrons of a **σ**-bond (usually **C-H** or **C-C**) with the adjacent, empty non-bonding p-orbital or an anti-bonding **σ*** or **π***-orbitals resulting in an extended molecular orbital. Unlike electromeric effect, hyper conjugation is a permanent effect.

It requires an α-CH group or a lone pair on atom like N, O adjacent to a **π** bond (sp^2 hybrid carbon). It occurs by the overlapping of the σ-bonding orbital or the orbital containing a lone pair with the adjacent π-orbital or p-orbital.

Example 1:

In propene, the *σ*-electrons of C-H bond of methyl group can be delocalised into the *π*-orbital of doubly bonded carbon as represented below.

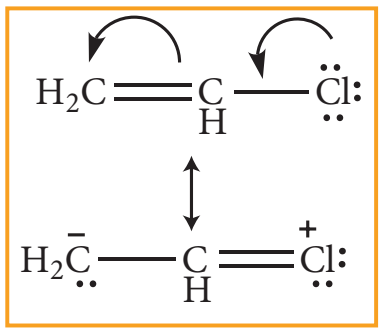


No bond resonance structures shown by propene are due to hyperconjugation

In the above structure the sigma bond is involved in resonance and breaks in order to supply electrons for delocalisation giving rise to 3 new canonical forms. In the contributing canonical structures: (II), (III) & (IV) of propene, there is no bond between an α-carbon and one of the hydrogen atoms. Hence the hyperconjugation is also known as "no bond resonance" or "Baker-Nathan effect". The structures (II), (III) & (IV) are polar in nature.

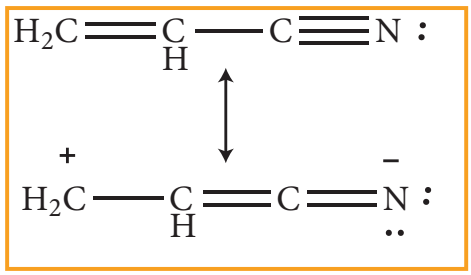
Example 2:

Hyper conjugation effect is also observed when atoms / groups having lone pair of electrons are attached by a single bond, and in conjugation with a **π bond**. The lone pair of electrons enters into resonance and displaces **π** electrons resulting in more than one structure



Example 3:

When electronegative atoms or group of atoms are in conjugation with a **π** -bond,they pull **π** - electrons from the multiple bond.



In case of carbocations, greater the number of alkyl groups attached to the